

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - FEB / MARCH 2025

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Software Engineering	Max. Marks	: 100
Course Code	: IS51 (2021,2022 Batch)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- With a neat diagram explain waterfall model and identify its advantages and its limitations. CO1 (08)
 - Compare and contrast plan driven processes and agile processes with a suitable use case. CO1 (07)
 - List out different ethical responsibilities of software engineering profession. CO1 (05)
- State and explain four different phases of Rational Unified Process (RUP) with a neat diagram. CO1 (06)
 - Analyze the advantages and disadvantages of the Rational Unified Process compared to agile methods. CO1 (08)
 - What is the need of extreme programming? Draw and describe the extreme programming cycle. CO1 (06)

UNIT - II

- What is Ethnography? Illustrate with a diagram how combining Ethnography and Prototyping helps the requirements discovery process CO2 (07)
 - Identify functional and non-functional requirements for an online e-commerce platform. CO2 (06)
 - What is requirement validation? List and explain any four different checks to be carried out during requirement validation Process. CO2 (07)
- What are the problems in natural language for specifying systems requirements? Explain how structured specifications can be used to overcome it. CO2 (07)
 - Develop a list of non-functional requirements for a real-time banking application CO2 (06)
 - Explain the structure of a Software Requirement Specification(SRS) document and highlight the characteristics of a good SRS document. CO2 (07)

UNIT - III

- Discuss in brief on how to effectively document the architectural design. CO3 (05)
 - What do you understand by the term "Architectural Style"? Discuss the different types of Architectural Styles with a suitable example. CO3 (10)
 - Discuss the advantages of explicitly designing and documenting a software architecture. CO3 (05)
- Discuss the different architecture models to be developed as a product in architectural design document. CO3 (05)
 - Justify the statement "A software system must be loosely coupled and highly cohesive". CO3 (05)

- c) Consider a **Car Rental System** case study where users can search for available cars, book rentals, and make payments online. The system also allows administrators to manage car inventory and view rental reports. CO3 (10)
- i. Design a **Use Case Diagram** to represent the system's functionality.
 - ii. Create a **Class Diagram** showing key classes, their attributes, methods, and relationships (e.g., association, aggregation, inheritance).
 - iii. Explain how Object-Oriented Design principles are applied in the design of the Car Rental System using the diagrams you created.

UNIT- IV

7. a) Discuss equivalence class test cases for the triangle problem. CO4 (10)
b) Differentiate between the specification –based and code-based testing. CO4 (10)
8. a) Explain the Robust and worst - case test cases for a function of two variables, in boundary value testing. CO4 (10)
b) Develop a formula for the member of robust worst - case test cases for a function of n variables. CO4 (10)

UNIT - V

9. a) What are the important goals of project management? Differentiate software engineering with other types of engineering. CO5 (10)
b) Distinguish between plan driven development and agile planning with a suitable example. CO5 (10)
10. a) List out three categories of risks with an example and explain the stages of risk management with diagram. CO5 (10)
b) Illustrate human needs hierarchy and explain three important needs that are most important from a management point of view. CO5 (05)
c) Explain software Pricing. List and briefly explain the factors affecting the software pricing. CO5 (05)
